

**PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE-MEC-I (B)**

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

Test Booklet Code

This Booklet contains **28** pages.**E5**

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject. **Candidates are advised to read all 15 questions in each subject of Section B** before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
- No candidate, without special permission of the Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

TG: @Chalnaayaaar

Your Target is to secure Good Rank in Pre-Medical 2024

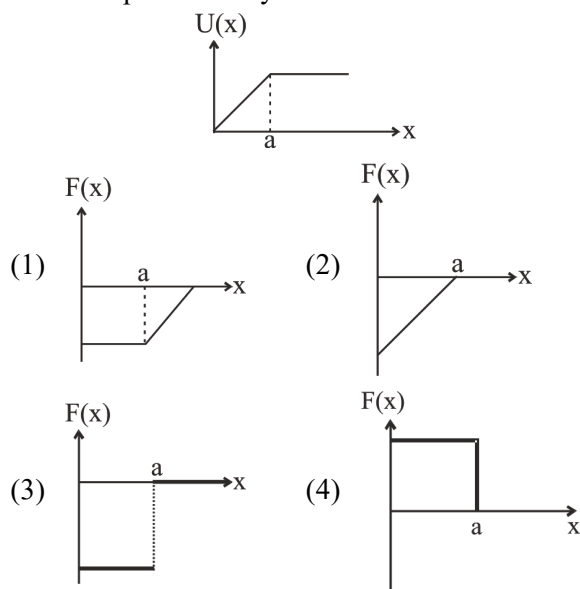
Topic : MAGNETIC FIELD, WORK, ENERGY AND POWER, WAVE OPTICS (ONLY POLARISATION), MODERN PHYSICS, [BACK UNIT] : RAY OPTICS AND OPTICAL INSTRUMENTS

SECTION - A (PHYSICS)

1. A man pulls a bucket of water from a depth of h from a well. If the mass of the rope and that of bucket full of water are m and M respectively, the work done by the man is :

- (1) $(M + m)gh$ (2) $\left(M + \frac{m}{2}\right)gh$
 (3) $\left(\frac{M + m}{2}\right)gh$ (4) $\left(\frac{M}{2} + m\right)gh$

2. The potential energy of a system is represented in the first figure, the force acting on the system will be represented by



3. A body of mass m is moved from rest, along a straight line, by an engine delivering constant power P . The velocity of the body after time t will be

- (1) $\sqrt{\frac{2Pt}{m}}$ (2) $\frac{2Pt}{m}$
 (3) $\sqrt{\frac{Pt}{2m}}$ (4) $\frac{Pt}{2m}$

4. A machine of 72% efficiency, uses 50 joule of energy from source to lift a 1kg mass through a certain distance. Mass is released from that distance. The velocity at ground will be

- (1) 6.5 m/sec (2) 7.5 m/sec
 (3) 8.5 m/sec (4) 9.5 m/sec

5. 50 N force is applied in direction of $3\mathbf{i} + 4\mathbf{j}$ so that object displaced from $(1,2,3)$ to $(3,1,2)$ find work done

- (1) 20 J (2) 100 J (3) 50 J (4) 0

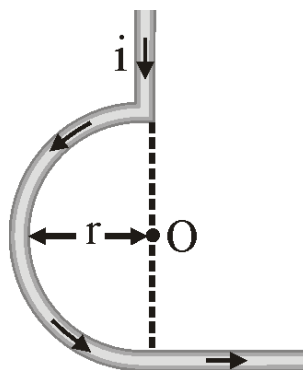
6. A body is released from height h . then ratio of its kinetic & potential energy after it travels $h/6$ distance

- (1) 1 : 5 (2) 1 : 4
 (3) 1 : 2 (4) 1 : 1

7. One metre length of wire carries a constant current. The wire is bent to form a circular loop. The magnetic field at the centre of this loop is B . The same is now bent to form a circular loop of smaller radius to have four turns in the loop. The magnetic field at the centre of this new loop is

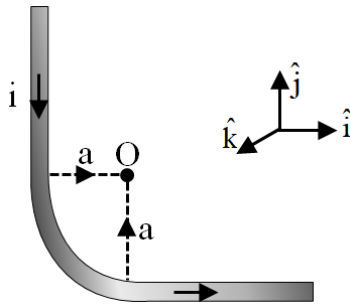
- (1) 4 B
 (2) 16 B
 (3) $\frac{B}{2}$
 (4) $\frac{B}{4}$

8. In the figure, what is the magnetic field at the point O

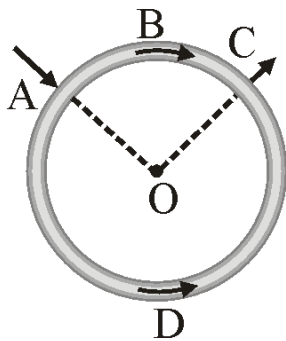


- (1) $\frac{\mu_0 I}{4\pi r}$ (2) $\frac{\mu_0 I}{4\pi r} + \frac{\mu_0 I}{2\pi r}$
 (3) $\frac{\mu_0 I}{4r} + \frac{\mu_0 I}{4\pi r}$ (4) $\frac{\mu_0 I}{4r} - \frac{\mu_0 I}{4\pi r}$

9. The unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$ are as shown below. What will be the magnetic field at O in the following figure.



- (1) $\frac{\mu_0}{4\pi} \frac{i}{a} \left(2 - \frac{\pi}{2}\right) \hat{j}$ (2) $\frac{\mu_0}{4\pi} \frac{i}{a} \left(2 + \frac{\pi}{2}\right) \hat{j}$
 (3) $\frac{\mu_0}{4\pi} \frac{i}{a} \left(2 + \frac{\pi}{2}\right) \hat{i}$ (4) $\frac{\mu_0}{4\pi} \frac{i}{a} \left(2 + \frac{\pi}{2}\right) \hat{k}$
10. Two similar coils are kept mutually perpendicular such that their centres coincide. At the centre, find the ratio of the magnetic field due to one coil and the resultant magnetic field by both coils, if same current is flow in both coils :
- (1) $1 : \sqrt{2}$ (2) $1 : 2$
 (3) $2 : 1$ (4) $\sqrt{3} : 1$
11. A uniform wire is bent in the form of a circle of radius R. A current I enters at A and leaves at C as shown in the figure: If the length ABC is half of the length ADC, the magnetic field at the centre O will be:

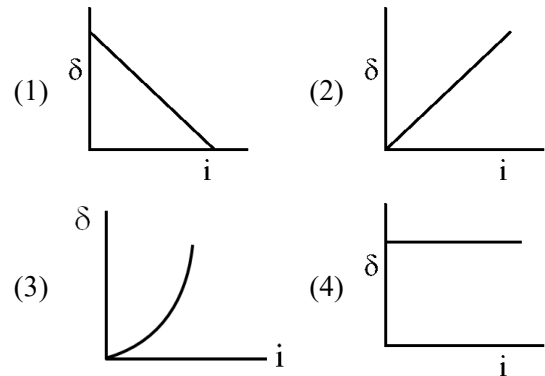


- (1) Zero (2) $\frac{\mu_0 I}{2R}$
 (3) $\frac{\mu_0 I}{4R}$ (4) $\frac{\mu_0 I}{6R}$

12. The field normal to the plane of a wire of n turns and radius r which carries a current i is measured on the axis of the coil at a small distance h from the centre of the coil. This is smaller than the field at the centre by the fraction :-

- (1) $\frac{3}{2} \frac{h^2}{r^2}$ (2) $\frac{2}{3} \frac{h^2}{r^2}$
 (3) $\frac{3}{2} \frac{r^2}{h^2}$ (4) $\frac{2}{3} \frac{r^2}{h^2}$

13. Graph between angle of deviation and incident angle in single reflection :-



14. A ray of light is incident on a plane mirror at an angle ' θ '. If deviation is 120° then θ will be :

- (1) 120°
 (2) 90°
 (3) 60°
 (4) 30°

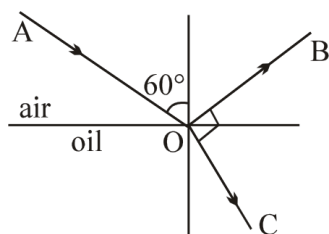
15. A virtual image three times the size of the object is obtained with a concave mirror of radius of curvature 36 cm. The distance of the object from the mirror is :-

- (1) 5 cm (2) 12 cm
 (3) 10 cm (4) 20 cm

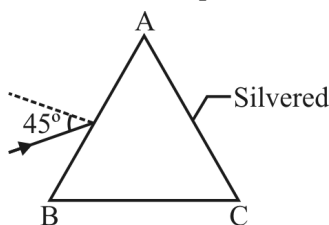
16. An air bubble is situated in glass slab of length 25 cm. When it is seen from one side, it appears at 6 cm and when seen from other side it appears at 4 cm. Find refractive index of slab :-

- (1) 1.5 (2) 2.5
 (3) 1 (4) 1.33

17. A ray of light AO is incident on the surface of oil. Reflected part of this ray OB and refracted part OC are mutually perpendicular as shown. Find refractive index of oil.



- (1) $\sqrt{3}$ (2) $\frac{2}{\sqrt{3}}$ (3) $\frac{1}{\sqrt{3}}$ (4) 2
18. A prism ABC of angle 30° has its face AC silvered. A ray of light incident at an angle of 45° at the face AB retraces its path after refraction at face AB and reflection at face AC. The refractive index of the material of the prism is :-



- (1) 1.5 (2) $\frac{3}{\sqrt{2}}$ (3) $\sqrt{2}$ (4) $\frac{4}{3}$
19. White light is incident on one of the refracting surfaces of a prism of angle 5° . If the refractive indices for red and blue colours are 1.641 and 1.659 respectively, the angular separation between these two colours when they emerge out of the prism is :-

- (1) 0.9° (2) 0.09° (3) 1.8° (4) 1.2°
20. **Assertion** :- Sky appears blue.
Reason :- Sensitivity of eye is higher for yellow colour as compare to violet colour.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
 (2) **Assertion** is true but **Reason** is false.
 (3) **Assertion** is false but **Reason** is true.
 (4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

21.

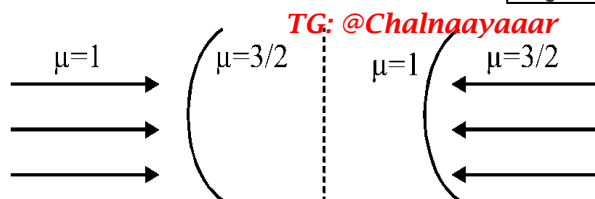


Fig.1

Fig.2

In given diagrams the lens will give rise to :-

- (1) In first figure $\rightarrow$ Convergent beam
 In second figure $\rightarrow$ Divergent beam
 (2) In first figure $\rightarrow$ Convergent beam
 In second figure $\rightarrow$ Convergent beam
 (3) In first figure $\rightarrow$ Divergent beam
 In second figure $\rightarrow$ Divergent beam
 (4) In first figure $\rightarrow$ Divergent beam
 In second figure $\rightarrow$ Convergent beam
22. The focal length of a convex lens is 10 cm and its refractive index is 1.5. If the radius of curvature of one surface is 7.5 cm, the radius of curvature of the second surface will be :-
- (1) 7.5 cm (2) 15.0 cm
 (3) 75 cm (4) 5.0 cm
23. An astronomical telescope has an angular magnification of magnitude 5 for distant objects. The separation between the objective and eye-piece is 36 cm and the final image is formed at infinity. Determine the focal length of objective and eye-piece—
- (1) $f_0 = 30$ cm, $f_e = 6$ cm
 (2) $f_0 = 25$ cm, $f_e = 10$ cm
 (3) $f_0 = 30$ cm, $f_e = 10$ cm
 (4) $f_0 = 15$ cm, $f_e = 5$ cm
24. At what angle should an unpolarised beam be incident on a crystal of $\mu = \sqrt{3}$, so that reflected beam is polarised :-
- (1) 45° (2) 60°
 (3) 90° (4) 0°

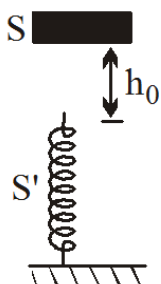
25. Let P and E denotes linear momentum and energy of photon. If the wavelength is decreased :-
 (1) Both P and E increase
 (2) P increase and E decrease
 (3) P decrease and E increase
 (4) Both P and E decrease
26. Wavelength of a neutron at temperature 27°C is λ . Its wavelength at 927°C will be ?
 (1) λ (2) 2λ (3) $\frac{\lambda}{2}$ (4) $\frac{\lambda}{4}$
27. The threshold wavelength for photoelectric emission in tungsten is 400 nm. The wavelength of light that must be used in order to eject electrons with a maximum energy of 0.9 eV to be :-
 (1) 120 nm (2) 310 nm
 (3) 380 nm (4) 400 nm
28. The electron in the hydrogen atom jumps from excited state ($n = 3$) to its ground state ($n = 1$) and the photons thus emitted irradiate a photosensitive material. If the work function of the material is 5.1 eV, the stopping potential is estimated to be (the energy of the electron in nth state $E_n = -\frac{13.6}{n^2}\text{eV}$):-
 (1) 12.1 V (2) 17.2 V
 (3) 7 V (4) 5.1 V
29. In Davisson-Germer experiment maximum intensity is observed at :-
 (1) 50° and 54 volt (2) 54° and 50 volt
 (3) 50° and 50 volt (4) 65° and 50 volt
30. According to the De Broglie, the radius of n^{th} Bohr's orbit is :-
 (λ = wavelength of electron in n^{th} orbit)
 (1) $\frac{n\lambda}{4}$ (2) $\frac{n\lambda}{2\pi}$
 (3) $n\lambda$ (4) $2n\lambda$
31. According to De-Broglie wavelength of electron in second orbit is 10^{-6} m. The circumference of orbit is :-
 (1) 10^{-6} m
 (2) 3×10^{-6} m
 (3) 4×10^{-6} m
 (4) 2×10^{-6} m
32. If De-broglie wavelength of a moving e^{-} is decreased from $1\text{\AA}$ to $0.5\text{\AA}$, then calculate change in its K.E. in eV :
 (1) 600 eV (2) 450 eV
 (3) 150 eV (4) 500 eV
33. 1 gm of hydrogen is converted in to 0.993 gm of helium in a thermonuclear reaction. The energy released is :
 (1) 63×10^7 J (2) 63×10^{10} J
 (3) 63×10^{14} J (4) 63×10^{20} J
34. If the radius of the ${}_{13}\text{Al}^{27}$ nucleus is taken to R_{Al} then the radius of ${}_{53}\text{Te}^{125}$ nucleus is nearly -
 (1) $\left(\frac{53}{13}\right)^{\frac{1}{3}} R_{\text{Al}}$
 (2) $\frac{5}{3} R_{\text{Al}}$
 (3) $\frac{3}{5} R_{\text{Al}}$
 (4) $\left(\frac{13}{53}\right)^{\frac{1}{3}} R_{\text{Al}}$
35. A nuclear fission is given below
 $\text{A}^{240} \rightarrow \text{B}^{100} + \text{C}^{140} + \text{Q}(\text{energy})$
 Let binding energy per nucleon of nucleus A, B and C is 7.6 MeV, 8.1 MeV and 8.1 MeV respectively. Value of Q is :-(Approximately)
 (1) 20 MeV (2) 220 MeV
 (3) 120 MeV (4) 240 MeV

SECTION - B (PHYSICS)

36. A uniform chain of length ℓ and mass m overhangs from a smooth table so that $\frac{2}{3}$ rd part of it is on the table then velocity of chain when it completely slips off the table :-

- (1) $\sqrt{2g\ell}$ (2) $\frac{2}{3}\sqrt{2g\ell}$
(3) $\sqrt{\frac{2}{3}g\ell}$ (4) None

37. A slab S of mass 1 kg is released from a height of 1cm from the top of a spring S' of force constant 780 N/m. The maximum compression x of the spring will be :



- (1) $\frac{2}{75}$ m (2) $\frac{1}{30}$ m
(3) $\frac{1}{78}$ m (4) None of these

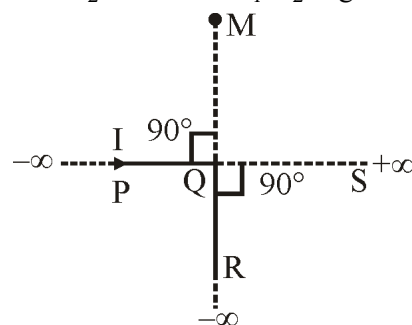
38. An engine pumps a liquid of density ρ through a pipe of cross section area A . If speed with which liquid passes is V . Then rate at which the kinetic energy is imparted on liquid is

- (1) $\frac{1}{2}A\rho v^3$ (2) $\frac{1}{2}\rho Av$
(3) $\frac{1}{2}\rho Av^2$ (4) ρAv^2

39. The ratio of the magnetic field at the centre of a current carrying circular wire and the magnetic field at the centre of a square coil made from the same length of wire will be

- (1) $\frac{\pi^2}{4\sqrt{2}}$ (2) $\frac{\pi^2}{8\sqrt{2}}$
(3) $\frac{\pi}{2\sqrt{2}}$ (4) $\frac{\pi}{4\sqrt{2}}$

40. An infinitely long conductor PQR is bent to form a right angle as shown. A current I flows through PQR, the magnetic field due to this current at the point M is H_1 . Now another infinity long straight conductor QS is connected at Q so that the current is $I/2$ in QR as well as in QS. The current in PQ remaining unchanged. The magnetic field at M is now H_2 . The ratio H_1/H_2 is given by

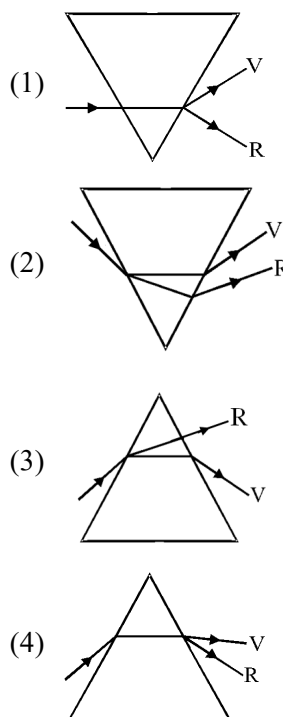


- (1) $\frac{1}{2}$ (2) 1 (3) $\frac{2}{3}$ (4) 2

41. A ray of light propagates from glass ($\mu_g = 3/2$) to water ($\mu_w = 4/3$). The value of the critical angle.

- (1) $\sin^{-1}(1/2)$ (2) $\sin^{-1}(\sqrt{8}/9)$
(3) $\sin^{-1}(8/9)$ (4) $\sin^{-1}(5/7)$

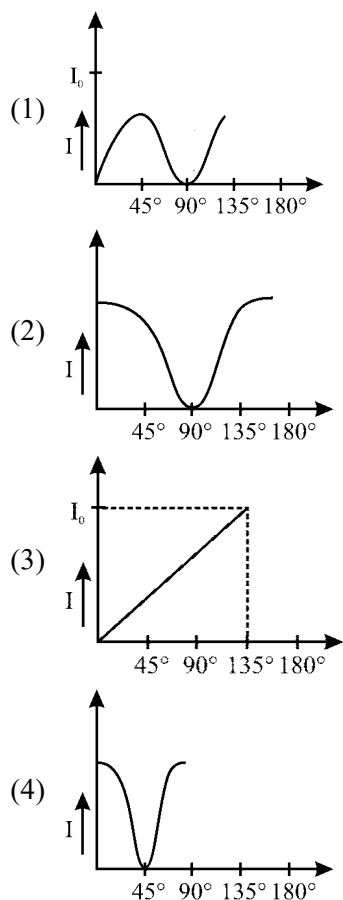
42. Which of the following diagrams shows correctly the dispersion of white light by a prism ?



43. In a compound microscope, the focal length of two lenses are 1.5 cm and 6.25 cm. If an object is placed at 2 cm from objective and the final image is formed at 25 cm from eye lens, the distance between two lenses is -

(1) 6.00 cm (2) 7.75 cm
(3) 9.25 cm (4) 11.0 cm

44. The graph showing the dependence of intensity of transmitted light on the angle between polariser and analyser, is :



45. Light of wavelength 5000 Å falling on a sensitive surface. If the surface has received 10^{-7} J of energy, then the number of photons falling on the surface will be -

(1) 5×10^{11}
(2) 2.5×10^{11}
(3) 3×10^{11}
(4) 1.6×10^{10}

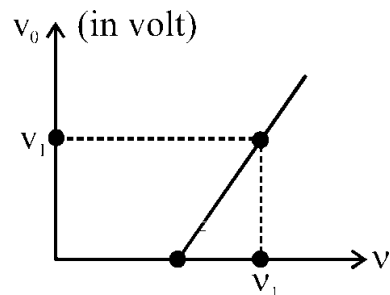
46. Electromagnetic wave of intensity 1400 W/m^2 falls on metal surface on area 1.5 m^2 is completely absorbed by it. Find out force exerted by beam :-

(1) $14 \times 10^{-5} \text{ N}$ (2) $14 \times 10^{-6} \text{ N}$
(3) $7 \times 10^{-5} \text{ N}$ (4) $7 \times 10^{-6} \text{ N}$

47. If the value of distance is 2m between source and cathode, the frequency of incident radiation is 10^{14} Hz then the photons per second is x now the distance becomes 4m and the frequency increases to $2 \times 10^{14} \text{ Hz}$ and power of source remain same then the value of photons per second (x') becomes :-

(1) $x' = \frac{x}{8}$ (2) $x' = \frac{x}{4}$
(3) $x' = 2x$ (4) $x' = 4x$

48. In P.E.E. obtain graph is :



Threshold frequency is :

(1) $\nu_1 - V_1$ (2) $\nu_1 + V_1$
(3) $\nu_1 - \frac{e}{h} V_1$ (4) $\nu_1 + \frac{e}{h} V_1$

49. The binding energies of the nuclei of ${}^4_2\text{He}$, ${}^7_3\text{Li}$, ${}^{12}_6\text{C}$ & ${}^{14}_7\text{N}$ are 28, 52, 90, 98 MeV respectively. Which of these is most stable.

(1) ${}^4_2\text{He}$ (2) ${}^7_3\text{Li}$ (3) ${}^{12}_6\text{C}$ (4) ${}^{14}_7\text{N}$

50. The binding energy of deuteron ${}^2_1\text{H}$ is 1.112 MeV per nucleon and an α -particle ${}^4_2\text{He}$ has a binding energy of 7.047 MeV per nucleon. Then in the fusion reaction ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^4_2\text{He} + Q$, the energy Q released is :-

(1) 1 MeV (2) 11.09 MeV
(3) 23.08 MeV (4) 931 MeV

Topic : GOC-II, HYDROCARBON, ALKYL HALIDE, OXYGEN CONTAINING COMPOUNDS, PERIODIC TABLE AND CHEMICAL BONDING, [BACK UNIT] : ELECTROCHEMISTRY

SECTION-A (CHEMISTRY)

51. E° for $\text{Mg}^{2+} / \text{Mg} = -2.37 \text{ V}$, $\text{Zn}^{2+} / \text{Zn} = -0.76 \text{ V}$ and $\text{Fe}^{2+} / \text{Fe} = -0.44 \text{ V}$.

Which statement is correct?

- (1) Zn reduces Fe^{2+}
- (2) Zn reduces Mg^{2+}
- (3) Mg oxidises Fe
- (4) Zn oxidises Fe

52. The standard EMF of a Daniel cell at 298 K is E_1 . When the concentration of ZnSO_4 is 1.0 M and that of CuSO_4 is 0.01 M, the EMF becomes E_2 at 298 K. The correct relationship between E_1 and E_2 is

- (1) $E_1 = E_2$
- (2) $E_2 = 0$
- (3) $E_1 > E_2$
- (4) $E_1 < E_2$

53. The oxidation potential of a hydrogen electrode at $\text{pH} = 1$ is ($T = 298 \text{ K}$) -

- (1) 0.059 volt
- (2) 0 volt
- (3) - 0.059 volt
- (4) 0.59 volt

54. Electrolysis of aqueous NaCl solution was carried out by passing 10 milli ampere current. The time required to liberate 0.01 mol of H_2 gas at the cathode is :-

- (1) $9.65 \times 10^4 \text{ sec}$
- (2) $19.3 \times 10^4 \text{ sec}$
- (3) $28.95 \times 10^4 \text{ sec}$
- (4) $38.6 \times 10^4 \text{ sec}$

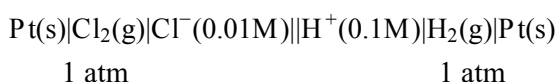
55. Given :

- (i) $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$, $E^\circ = 0.337 \text{ V}$
- (ii) $\text{Cu}^{2+} + \text{e}^- \rightarrow \text{Cu}^+$, $E^\circ = 0.153 \text{ V}$

Electrode potential, E° for the reaction, $\text{Cu}^+ + \text{e}^- \rightarrow \text{Cu}$, will be :-

- (1) 0.38 V
- (2) 0.52 V
- (3) 0.90 V
- (4) 0.30 V

56. E_{cell} for



$$E^\circ_{\text{Cl}_2/\text{Cl}^-} = 1.36 \text{ V}$$

- (1) -1.30 V
- (2) 1.54 V
- (3) -1.54 V
- (4) 1.30 V

57. Given $E^\circ_{\text{Cl}_2/\text{Cl}^-} = 1.36 \text{ V}$, $E^\circ_{\text{Cr}^{3+}/\text{Cr}} = -0.74 \text{ V}$

$$E^\circ_{\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}} = 1.33 \text{ V}, E^\circ_{\text{MnO}_4^-/\text{Mn}^{2+}} = 1.51 \text{ V}$$

Then select the incorrect statement -

- (1) Strongest oxidant among above given substances is MnO_4^-
- (2) Strongest reductant among above given substances is Cr^{+3}
- (3) Order of oxidising power is $\text{MnO}_4^- > \text{Cl}_2 > \text{Cr}_2\text{O}_7^{2-} > \text{Cr}^{+3}$
- (4) $\text{Cr}^{+3} | \text{Cr}$ redox couple act as anode with SHE to form galvanic cell which work spontaneously under standard conditions

58. In a galvanic cell, salt bridge -

- (A) Does not participate chemically in cell reaction.
 - (B) Maintain electrical neutrality of both electrolytic solution.
 - (C) Ensures mixing of two electrolytic solution.
 - (D) Is necessary for the occurrence of cell reaction
- From above statements which are correct-

- (1) A, B and C
- (2) A, B and D
- (3) Only B
- (4) Only B and D

TG: @Chalnaagyaar

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59. During electrolysis of 2M NaCl (aq.) solution with Pt anode and Hg cathode, products obtained at anode and cathode respectively are -

- (1) Cl_2, H_2
- (2) H_2, Cl_2
- (3) Na - Hg amalgam, Cl_2
- (4) Cl_2 , Na - Hg amalgam

60. Incorrect statement regarding fuel cells -

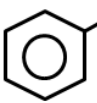
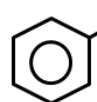
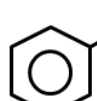
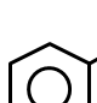
- (1) These cells are designed to convert energy of combustion of fuels like hydrogen, methane, methanol directly into electrical energy.
- (2) They produce electricity with an efficiency of about 40% compared to thermal plants whose efficiency is about 70 %
- (3) It is a type of galvanic cell
- (4) In hydrogen fuel cell, overall reaction is $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\ell)$

61. **Assertion** : Lucas reagent is a mixture of anhydrous ZnCl_2 and concentrated HCl.

Reason : Primary alcohol produce precipitate with Lucas reagents with in few second.

- (1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- (2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- (3) Assertion is true but Reason is false.
- (4) Both Assertion and Reason are false.

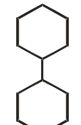
62. Best pair to prepare Anisole is by using Williamson ether synthesis is ?

- (1)  + $\text{CH}_3\text{O}^-\text{Na}^+$
- (2)  + $\text{CH}_3\text{O}^-\text{Na}^+$
- (3)  + $\text{CH}_3 - \text{I}$
- (4)  + CH_3F

63. Which of the following sequence of reactions is suitable to prepare methane

- (1) $\text{CH}_3\text{COONa} \xrightarrow{\text{electrolysis}}$
- (2) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{H} \xrightarrow[\text{HCl}]{\text{Zn-Hg}}$
- (3) $\text{CH}_3\text{COONa} \xrightarrow{\text{NaOH} + \text{CaO}}$
- (4) $\text{CH}_3 - \text{Br} \xrightarrow[\text{dry ether}]{\text{Na}}$

64.  $\xrightarrow[\text{h}\nu]{\text{Br}_2}$ (A) $\xrightarrow[\text{Dry ether}]{\text{Na}}$ (B)
Product B is :-

- (1) 
- (2) 
- (3) 
- (4) 

65. $\text{CH}_3 - \text{CH}_2 - \text{CH} = \text{CH}_2 \xrightarrow{\text{HBr}} \text{A} \xrightarrow{\text{alc. KOH}} \text{B (major)}$
 $\downarrow \begin{matrix} \text{(i) O}_3 \\ \text{(ii) Zn/H}_2\text{O} \end{matrix}$
 C

Product 'C' is :-

- (1) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{COOH}$
- (2) $\text{CH}_3 - \text{CHO}$
- (3) $\text{CH}_3 - \text{CHO} + \text{CH}_3 - \text{CH}_2 - \text{CHO}$
- (4) $\text{CH}_3 - \text{COOH}$

66. Match the column :

Column-I		Column-II	
(a)	Zn-Hg+HCl	(i)	Rosenmund catalyst
(b)	H ₂ +Pd-BaSO ₄	(ii)	Clemmensen reagent
(c)	NH ₂ -NH ₂ + [⊖] OH ⁺ ethylene glycol	(iii)	Birch reagent
(d)	Na+liquid NH ₃	(iv)	wolff kishner reagent

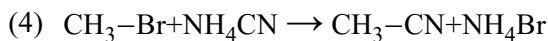
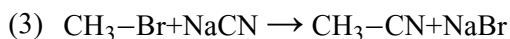
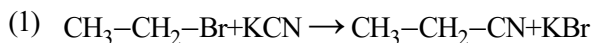
(1) a-(i), b-(ii), c-(iii), d-(iv)

(2) a-(ii), b-(i), c-(iv), d-(iii)

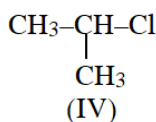
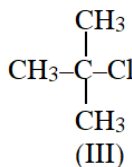
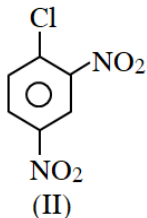
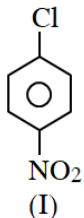
(3) a-(ii), b-(i), c-(iii), d-(iv)

(4) a-(i), b-(ii), c-(iv), d-(iii)

67. Incorrect product is :-



68. Correct order of reactivity towards hydrolysis is :-



(1) I > II > III > IV

(2) IV > III > II > I

(3) II > I > III > IV

(4) III > IV > II > I

69. Which of the following will give immediate turbidity with Lucas reagent :

(1) Isobutyl alcohol

(2) 2-methyl butan-3-ol

(3) 2-Methyl butan-2-ol

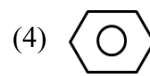
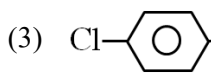
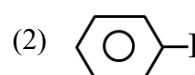
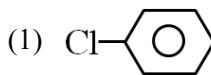
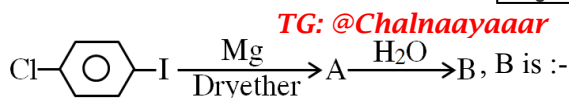
(4) Butan-2-ol

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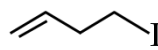
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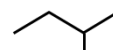
70.



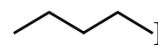
71. Correct order of reactivity of dehydrohalogenation is :-



(a)



(b)



(c)

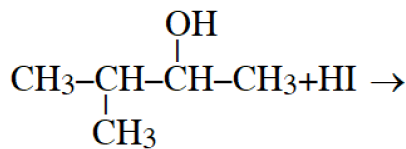
(1) c > b > a

(2) a > c > b

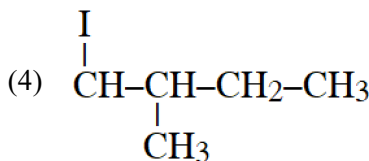
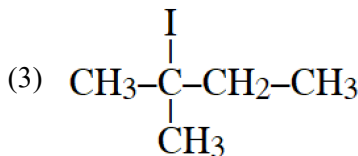
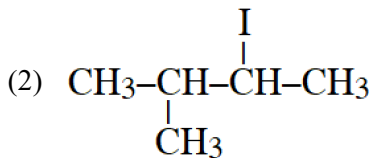
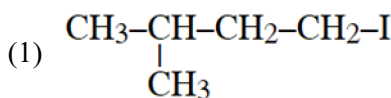
(3) a > b > c

(4) c > a > b

72.

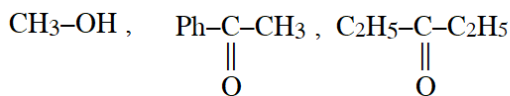
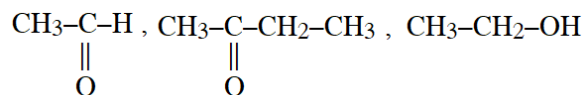


Major product is



73.

How many of following give haloform reaction.



(1) 6

(2) 3

(3) 5

(4) 4

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74. Which of the following is smallest in size :-
 (1) $1s^2, 2s^2 2p^6, 3s^1$
 (2) $1s^2, 2s^2 2p^6, 3s^2 3p^5$
 (3) $1s^2, 2s^2 2p^6, 3s^2 3p^3$
 (4) $1s^2, 2s^2 2p^6, 3s^2$
75. Consider the given elements /ion, Mg, Mg^{2+} , Al, Al^{3+} . Find species which has largest & smallest radius respectively :-
 (1) Al & Al^{3+}
 (2) Mg & Mg^{2+}
 (3) Mg & Al^{3+}
 (4) Al & Mg^{2+}
76. Incorrect order of electron affinity is
 (1) $S > O$
 (2) $O > Se$
 (3) $Cl > F$
 (4) $Li > B$
77. The electron gain enthalpies (in kJ mol^{-1}) of three halogens, X, Y and Z are respectively -349, -333 and -325. Then X, Y and Z are respectively :-
 (1) F, Cl and Br
 (2) Cl, F and Br
 (3) Cl, Br and F
 (4) Br, Cl and F
78. Which of the following set of molecules doesn't follow octet rule ?
 (1) CO_2, PCl_3, CF_4
 (2) CO_2, NH_3, H_2O
 (3) $AlCl_3, BF_3, PCl_5, SF_6$
 (4) H_2O, NH_3, CO_2, SF_2
79. The geometry and hybridisation of XeO_2F_2 -
 (1) Square planar, sp^3d
 (2) tetrahedral, sp^3
 (3) see-saw, sp^3d
 (4) octahedral, sp^3d^2
80. Which one is isostructural pairs :
 (1) NF_3 and BF_3
 (2) BF_4^- and NH_4^+
 (3) BCl_3 and BrF_3
 (4) NH_3 and BF_3
81. Which one have diagonal hybridisation :
 (1) BeF_2 (2) CF_4 (3) SF_4 (4) PF_5
82. Which of the following have more than one type of bond :
 (1) SF_6 (2) SiF_4
 (3) C_2F_6 (4) $BeCl_2$
83. Which pair is amphoteric :
 (1) MnO and BeO
 (2) Cr_2O_3 and MnO_2
 (3) MgO and ZnO
 (4) CrO_3 and V_2O_5
84. Which can have covalency more than four :
 (1) Be
 (2) B
 (3) N
 (4) P
85. Number of Lone pair at central atom for $XeOF_2$.
 (1) 2
 (2) 3
 (3) 1
 (4) 4

SECTION-B (CHEMISTRY)

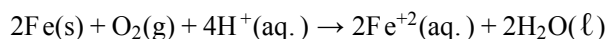
86. Resistance of 0.2 M solution of an electrolyte placed in conductivity cell is 5Ω . The specific conductance of the solution is 1.4 S m^{-1} . The resistance of 0.5 M solution of the same electrolyte placed in same cell is 280Ω . The molar conductivity of 0.5 M solution of the electrolyte in $\text{S m}^2 \text{ mol}^{-1}$ is :

- (1) 5×10^3 (2) 5×10^2
(3) 5×10^{-5} (4) 5×10^{-3}

87. A certain current liberates 0.500 g of H_2 in 2.00 hours. How many gram of oxygen can be liberated by the same current in the same time?

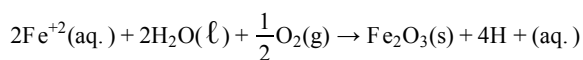
- (1) 0.500 g (2) 8.00 g
(3) 4.00 g (4) 16.00 g

88. **Assertion (A) :-** Overall reaction during rusting of iron is



$E^\circ_{\text{cell}} = 1.67 \text{ V}$ since E°_{cell} is positive, cell reaction is non spontaneous.

Reason (R) :- The ferrous ions are further oxidised by atmospheric oxygen to ferric ion which comes out as rust in the form of hydrated ferric oxides.



- (1) Both A and R are true but R is not correct explanation of A.
(2) A is true but R is false
(3) A is false but R is true
(4) Both A and R are true and R is correct explanation of A.

TG: @Chalnaayaaar

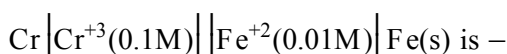
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89. For the reduction of silver ions with copper metal, the standard cell potential was found to be $+0.46\text{V}$ at 25°C . The value (approximate) of standard Gibbs energy, ΔG° will be ($F = 96500 \text{ C mol}^{-1}$)

- (1) -98.0 kJ
(2) -89.0 kJ
(3) $+89.0 \text{ kJ}$
(4) -44.5 kJ

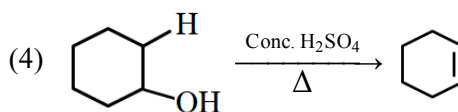
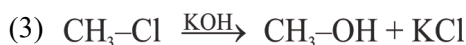
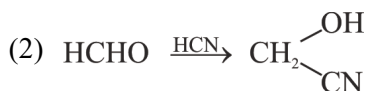
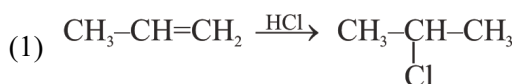
90. Potential of cell



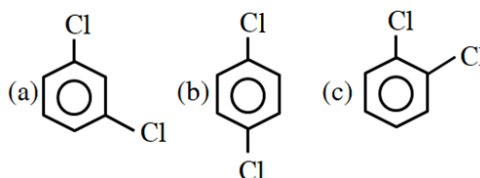
(Given $E^\circ_{\text{Fe}^{+2}/\text{Fe}} = -0.42\text{V}$, $E^\circ_{\text{Cr}^{+3}/\text{Cr}} = -0.72\text{V}$)

- (1) -0.34V
(2) -0.26V
(3) $+0.26\text{V}$
(4) 0.34V

91. Which of the following is an example of substitution reaction :-



92. Select correct order for :

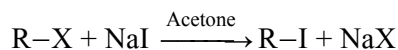


- (1) Boiling point $a > b > c$
(2) Melting point $c > b > a$
(3) Boiling point $c > b > a$
(4) Melting point $b > a > c$

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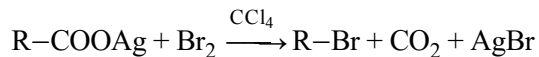
93. Incorrect match is :

(1) Finkelstein reaction



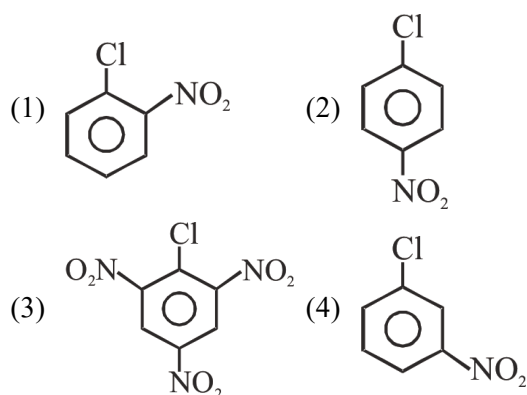
(2) Swarts reaction $R-X + Ag-F \rightarrow R-F + AgX$

(3) Darzen process

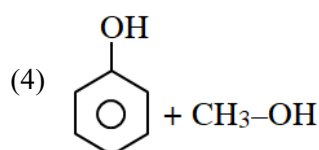
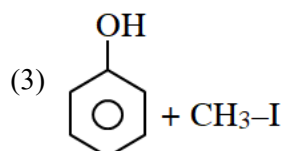
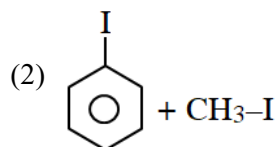
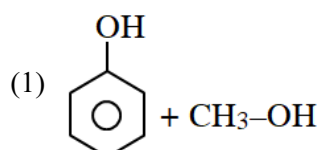
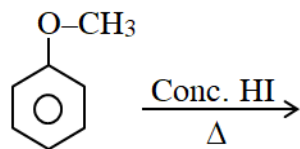


(4) None

94. Which of the following is least reactive towards S_NAr :-



95. Write down major product :



96. For the elements B, C, N, F & Si, the correct order of their non-metallic character is :-

(1) $B > C > Si > N > F$

(2) $Si > C > B > N > F$

(3) $F > N > C > B > Si$

(4) $F > N > C > Si > B$

97. Which of the following order is correct -

(1) Basic character : $Li_2O > Na_2O > K_2O > Cs_2O$

(2) Acidic character : $Al_2O_3 < SiO_2 < P_2O_5 < Cl_2O_7$

(3) Acidic strength : $HF > HCl > HBr > HI$

(4) Acidic character : $N_2O > NO_2 > N_2O_5 > N_2O_3$

98. In which of the following d-orbitals are not used by central atom in hybridization ?

(1) PF_5

(2) PCl_5

(3) XeO_3

(4) XeF_6

99. Which of the following have same number of sigma and Pi bond :

(1) Ethene

(2) Water

(3) Benzene

(4) Carbon suboxide

100. **Statement-I** : Nitrogen exist in the form of diatomic molecule but phosphorous in tetraatomic.

Statement-II : Phosphorous do not have d-orbital.

(1) Both statement are correct.

(2) Statement-I is correct but statement-II is incorrect.

(3) Both statement are incorrect.

(4) Statement-I is incorrect but statement-II is correct.

Topic : DIGESTIVE SYSTEM, CIRCULATORY SYSTEM, PRINCIPLES OF INHERITANCE (RECIPROCAL CROSS TO LAST), [BACK UNIT] : ECOLOGY

TG: @Chalnaayaaar

SECTION-A (BIOLOGY-I)

101. According to the evolutionary biologists the success of mammals is due to their :

- (1) Ability of reproduction
- (2) Ability to maintain a constant body temperature
- (3) Migrate or suspend in wake of unfavourable conditions
- (4) All of the above

102. **Assertion:** Gaseous exchanges of phosphorus between organism and environment are negligible.

Reason: Amount of phosphorus in atmosphere is negligible and organisms do not use phosphorus in respiration.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

103. **Assertion :** No energy that is trapped into an organism remains in it forever.

Reason : The energy trapped by the producer is either passed to a consumer or to decomposers when it dies.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

104. **Statement-I :** The pyramid of biomass in sea is generally upright.

Statement-II : Pyramid of energy is always upright.

- (1) Statement-I & Statement-II both are correct.
- (2) Statement-I is correct & Statement-II is incorrect.
- (3) Statement-I is incorrect & Statement-II is correct.
- (4) Both Statement-I & Statement-II is incorrect.

105. **Statement-I :** The three steps in the decomposition fragmentation, leaching and catabolism operate on the detritus simultaneously.

Statement-II : Humification and mineralisation occurs during decomposition in the soil.

- (1) Statement-I & Statement-II both are correct.
- (2) Statement-I is correct & Statement-II is incorrect.
- (3) Statement-I is incorrect & Statement-II is correct.
- (4) Both Statement-I & Statement-II is incorrect.

106. Blood group AB means :-

- (1) Antigen A and B in plasma and no antibody on R.B.C.
- (2) Antibody a and b on R.B.C. and no antigen in plasma.
- (3) Antigen A and B on R.B.C. and no antibody a and b in plasma.
- (4) Antigen A on R.B.C. and antigen B in plasma.

107. A child is of blood group 'O'. His parents blood group can not be :-

- (1) B & O
- (2) A & O
- (3) AB
- (4) A & B

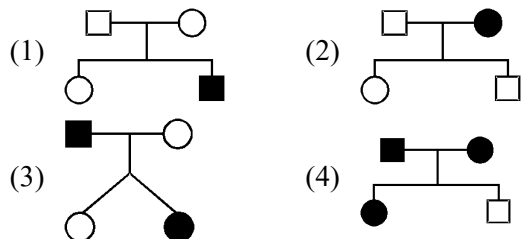
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108. If 7 alleles are responsible for a character than total possible genotypes of this character will be –
 (1) 28 (2) 10 (3) 10 (4) 15
109. A man with blood group B marries a female with blood group A and their first child is having blood group B. What is the genotype of child ?
 (1) $I^A I^B$ (2) $I^A I^O$
 (3) $I^B I^O$ (4) $I^B I^B$
110. Occasionally, a single gene may express more than one effect. This is :
 (1) Polygenic inheritance
 (2) Pleiotropy
 (3) Multiple allelism
 (4) Co-dominance
111. Human skin colour is controlled by 3 genes. which genotype show darkest & lightest skin colour respectively :-
 (1) AABBCc and aabbcc
 (2) aabbcc and AABBCc
 (3) AaBbCc and AaBbCc
 (4) AABBCc and aabbcc
112. Only parental combinations are formed, if :-
 (1) Genes show independent assortment
 (2) Genes are completely linked on chromosome
 (3) Genes are incompletely linked on chromosomes
 (4) Crossing over is maximum
113. If two colourblind persons get married, what is the probability of their first daughter being colourblind ?
 (1) 25%
 (2) 50%
 (3) 100%
 (4) zero
114. A colourblind man has a colourblind sister, but a normal brother, then his mother would be -
 (1) Colourblind
 (2) Completely normal
 (3) Normal but carrier
 (4) May be carrier or colourblind
115. Which statement is true for sex determination in honey-bee ?
 (1) Sex is determined by sex index ratio
 (2) Female is formed by means of parthenogenesis
 (3) Females are diploid having 32 chromosomes
 (4) Male (Drone) produce sperm by meiosis
116. Male heterogamety is found in all of these except :
 (1) Human beings
 (2) Birds (fowl)
 (3) *Drosophila*
 (4) Grass hopper
117. Males produce sperm by mitosis in :-
 (1) Drone in honey bee
 (2) Man
 (3) Reptiles
 (4) Plants
118. Which of the following pedigree chart can not be for sickle cell anaemia ?



119. **Assertion** : In Morgan's experiment cross A shows few recombinants (1.3%) in F_2 generation.

Reason : In cross A, genes of body color and eye color are tightly linked on X-chromosome.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

120. **Assertion (A)** : AB blood group is example of co-dominance.

Reason (R) : I^A & I^B when present together, they both express their own type of sugar.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

121. **Statement-I** : Colour blindness is autosomal recessive disorder.

Statement-II : Colour blindness occur in about 8% of males and only about 0.4% of females.

- (1) Both Statement are true
- (2) Statement-II is incorrect statement-I correct
- (3) Statement-I is incorrect and statement-II is correct
- (4) Both Statement are incorrect

122. Read the following statements and choose the correct option.

Statement-I : Males generally function as carriers of sex linked disorders.

Statement-II : Mother passes the alleles of a sex linked trait to daughters only.

- (1) Both the statements are correct.
- (2) Only statement-I is correct
- (3) Only statement-II is correct
- (4) Both the statements are incorrect.

123. Human liver fluke depends on two intermediate hosts to complete its life cycle, they are :-

- (1) Snail and Frog
- (2) Snake and Fish
- (3) Snail and Fish
- (4) Snake and Frog

124. Which of the following ecologist described that five closely related species of warblers living on the same tree were able to avoid competition and co-exist due to behavioural differences in their foraging activities ?

- (1) Connel
- (2) Mac Arthur
- (3) Ramdeo Mishra
- (4) Arthur Tansley

125. **Statement-I** : But for predators, prey populations could achieve high numbers.

Statement-II : Predators are too efficient for predation.

- (1) Statement-I & Statement-II both are correct.
- (2) Statement-I is correct & Statement-II is incorrect.
- (3) Statement-I is incorrect & Statement-II is correct.
- (4) Both Statement-I & Statement-II is incorrect.

126. A species whose distributions restricted to a small geographical area because of the presence of a competitively superior species, is found to expand its distributional range dramatically when the competing species is experimentally removed. This is the observation of :-
- (1) Resource partitioning
 - (2) Competitive release
 - (3) Interference competition
 - (4) Competitive exclusion
127. Extinct bird dodo was native of which of the following place?
- (1) Maldives
 - (2) Australia
 - (3) Mauritius
 - (4) Africa
128. How many of following would come under bioprospecting ?
Firewood, medicines, pollination, pest control, Fibre, Fruits, oxygen
- (1) 3 (2) 4
 - (3) 2 (4) 1
129. Which of the following is not an example of hotspot in Indian sub-continent?
- (1) Western Ghats and Srilanka
 - (2) Indo-Burma
 - (3) Himalaya
 - (4) Eastern Ghats
130. The historic convention on biological diversity was held in :-
- (1) Rio de Janerio in 1992
 - (2) Johanesburg in 2002
 - (3) Montreal in 1987
 - (4) Kyoto in 1992
131. Which are major cause of atmospheric pollution in metrocities?
- (1) Automobiles
 - (2) Industries
 - (3) Thermal power plants
 - (4) Burning waste in open areas
132. Which of the following inactivates catalyst in catalytic converter?
- (1) SO₂
 - (2) CO
 - (3) Pb
 - (4) SPM (Suspended particulate matter)
133. Which is he world's most problematic aquatic weed found in eutrophic lakes?
- (1) *Nelumbo* (2) *Nymphae*
 - (3) *Eichhornia* (4) *Vallisneria*
134. Which of the following is NOT water pollutant ?
- (1) DDT
 - (2) Mercury
 - (3) Thermal waste water
 - (4) Carbon dioxide
135. How many of the following come under the broad utilitarian role of ecosystem?
Pollination, pest control, Food, Fibre, Medicines, Climate moderation, Flood control
- (1) 4 (2) 5 (3) 6 (4) 2

SECTION-B (BIOLOGY-I)

136. Benthic organisms are affected most by :-
- (1) Light reaching the ocean floor
 - (2) Surface turbulence of water
 - (3) Sediment characteristics of aquatic ecosystems
 - (4) Water - holding capacity of soil

137. Out of the total cost of various ecosystem services, alone soil formation costs upto 50 percent. How much would climate regulation and habitat for wildlife cost upto ?

- (1) 5 percent each
- (2) 6 percent each
- (3) 4 percent each
- (4) 10 percent each

138. Possible blood group in children from the parents with 'B' and 'O' blood group are :-

- (1) All 'B'
- (2) All 'O'
- (3) Both 'B' & 'O'
- (4) 'A' & 'B'

139. How many phenotypes are possible in F_2 generation if a character is controlled by 5 polygenes :-

- (1) 32
- (2) 11
- (3) 243
- (4) 81

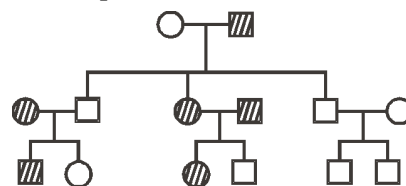
140. The minimum weight of tomatoes obtained from a plant is 40g. The maximum weight of tomatoes obtained is 140g. What will be the weight of tomatoes obtained by crossing plants with genotype $AABB \times aabb$? Fruit weight is controlled by 2 genes.

- (1) 50g
- (2) 70g
- (3) 90g
- (4) 25g

141. Maximum height of a plant is 30ft. and minimum height is 10 ft. If plant height is controlled by 3 pairs of genes, then the height of plant with genotype $AaBbCc$ will be -

- (1) 13.2 ft.
- (2) 23.2 ft.
- (3) 16.6 ft.
- (4) 19.9 ft.

142. Study the given pedigree carefully and identify the inheritance pattern ?



- (1) X-linked dominant
- (2) X-linked recessive
- (3) Autosomal recessive
- (4) Cytoplasmic inheritance

143. In the rocky intertidal communities of American pacific coast, when all the starfishes were removed than more than 10 species of invertebrates got extinct within a year. The main reason for this extinction was :

- (a) Exploitation
- (b) Interspecific competition
- (c) Intraspecific competition
- (d) Predation

- (1) b and d
- (2) a and d
- (3) a, b and c
- (4) only b

144. Match the column

A.	Xerarch succession	(i)	Red swamp stage
B.	Primary succession	(ii)	Stable as long as environment remains unchanged
C.	Climax community	(iii)	Slow process
D.	Hydrarch succession	(iv)	Occurs in dry areas

- (1) A - (i), B - (ii), C - (iii), D - (iv)
- (2) A - (i), B - (iii), C - (ii), D - (iv)
- (3) A - (iv), B - (iii), C - (ii), D - (i)
- (4) A - (iv), B - (iii), C - (i), D - (ii)

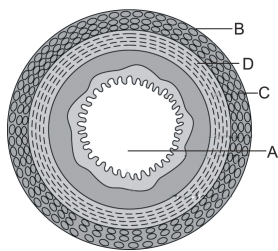
145. A study of a freshwater community revealed the following food chain:
- Phytoplankton → zooplankton → perch → trout → osprey
- The amount of energy in phytoplankton in this community is 1,000,000 joules. How much of this energy would most likely appear in the tertiary consumer given a 10% efficiency transfer between trophic levels?
- (1) 100,000 joules
(2) 10,000 joules
(3) 1,000 joules
(4) 100 joules
146. Which is/are correct from followings ?
- (A) Population ecology links ecology to population genetics and evolution.
(B) Births and deaths are features of population.
(C) If a new habitat is just being colonised, immigration may contribute more significantly to population growth than birth rate.
(D) The exponential growth model is considered a more realistic one.
- (1) A & B are correct
(2) B & C are correct
(3) C & D are correct
(4) A and C are correct
147. In a pond, there were 30 lotus plants in last year. Through reproduction, 25 new lotus plants were added by reproduction in one year while 8 plants died. The birth and death rate for the lotus population respectively are _____ and _____ individuals per lotus per year :-
- (1) 0.83, 0.26
(2) 0.26, 0.83
(3) 0.25, 0.80
(4) 0.80, 0.25
148. Which of the following is not a step taken by Delhi govt. to control vehicular pollution?
- (1) Phasing out old vehicles
(2) Use of unleaded & low sulphur petrol
(3) Use of catalytic converters
(4) Mixing 5% ethyl alcohol in petrol
149. How many of the following are colloidal impurities present in polluted water?
Sand, Bacteria, Cloth, Paperfibres, Sodium, Ammonia
- (1) 2
(2) 3
(3) 4
(4) 1
150. A town inhabited by a small group of population is surrounded by thermal power plant and odd number of factories, which of the following cases are observed in the town?
- (a) Soil is slightly acidic
(b) Outer surface of monuments got eroded
(c) Native flora and fauna of nearby lake started disappearing.
(d) Reduction in the population of lichens.
- (1) a, b
(2) b, c
(3) a, b, c
(4) a, b, c, d

SECTION-A (BIOLOGY-II)

151. Starch synthesis (starch branching enzyme) in seeds of garden pea is controlled by how many genes?
(1) 2 (2) 4 (3) 1 (4) 3
152. If more than one phenotype is influenced by same gene, then dominance depends on which of the following factors?
(1) Gene and the product that the gene has information.
(2) Production of particular phenotype from gene product.
(3) Particular phenotype we choose to examine
(4) All of these
153. Who united the knowledge of chromosomal segregation with Mendelian principles and called it as chromosomal theory of inheritance?
(1) Sutton (2) Boveri
(3) Morgan (4) Correns
154. Who experimentally verified chromosomal theory of inheritance?
(1) Morgan
(2) Sturtevant
(3) Mendel
(4) Sutton and Boveri
155. In 1891, Henking could trace a specific nuclear structure all through spermatogenesis in many insects. This specific structure was later called
(1) X-chromosome
(2) Y-chromosome
(3) Z-chromosome
(4) W-chromosome

156. In a particular species the males have one less chromosome than their female counterparts. What could be the name of that species?
(1) Human (2) Fruit Fly
(3) Garden Pea (4) Grasshopper
157. Which is responsible for determining sex of offsprings of fowls (chicks)?
(1) Egg
(2) Sperm
(3) Either Sperm or Egg
(4) Environment
158. They do not have father and son, but do have grand father and grandson. This statement is applicable on :-
(1) Worker Bee (2) Queen Bee
(3) Male Drone (4) Female Bee
159. Which of the following gave greater credibility to the data that Mendel collected in his experiments?
(1) Bisexuality of garden pea
(2) Large sampling size
(3) Qualitative characters
(4) Every character had two opposite/contrasting traits
160. What is the probability of gamete AB formed by Ab/aB if the distance between genes is 10 cM?
(1) 10% (2) 20% (3) 90% (4) 5%
161. Which of the following is example of pleiotrophy?
(1) Phenylketonuria
(2) Starch synthesis in garden pea
(3) ABO blood group
(4) Both 1 and 2

162.



Label the diagram.

- (1) A → Lumen, B → Serosa, C → Longitudinal Muscle, D → Circular Muscle
- (2) A → Longitudinal Muscle, B → Lumen, C → Serosa, D → Circular Muscle
- (3) A → Lumen, B → Longitudinal Muscle, C → Serosa, D → Circular Muscle
- (4) A → Longitudinal Muscle, B → Lumen, C → Circular Muscle, D → Serosa

163. Wrong about bile juice is :-

- (1) Emulsification of fats
- (2) Help in absorption of fat
- (3) Have lipase enzyme
- (4) Excretion of bilirubin and biliverdin

164. Read the following statements :-

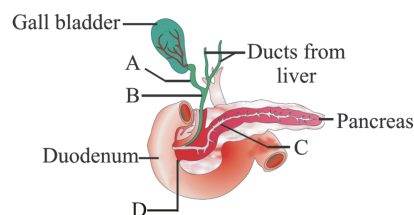
- (A) fatty acid & glycerol are absorbed through mucosa of small intestine into blood.
- (B) fatty acid & glycerol are absorbed through mucosa of small intestine into lymph.
- (C) Simple sugar and alcohol are absorbed in stomach.
- (D) Most of the absorption take place in small intestine.

From above statements which is/are **incorrect** ?

- (1) A and C
- (2) B and D
- (3) B and C
- (4) Only A

165.

The given figure is a duct system of liver, gall bladder and pancreas. Identify the names of ducts from A to D :-



- (1) A - Cystic duct, B - Pancreatic duct, C - Bile duct, D - Hepato - Pancreatic Duct
- (2) A - Cystic duct, B - Bile duct, C - Hepato - Pancreatic Duct, D - Pancreatic duct
- (3) A - Bile duct, B - Cystic duct, C - Pancreatic duct, D - Hepato - Pancreatic Duct
- (4) A - Cystic duct, B - Common Bile duct, C - Main Pancreatic duct, D - Hepato - Pancreatic Duct

166. **Assertion** : The stomach converts the food into chyme.

Reason : The food mixes thoroughly with the acidic gastric juice by the churning movements.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

167. **Assertion** : Human beings have 2 sets of teeth during their life.

Reason : Human beings have thecodont type of dentition.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

168. Read the following statements (A-C) and answer the question following them.

(A) chylomicrons are very small proteins coated fat globules.

(B) Defaecation is a voluntary process mediated by peristalsis.

(C) Inflammation of intestinal tract is the most common gut ailment.

How many of the above statements are correct?

- (1) None
- (2) One
- (3) Two
- (4) Three

169. **Statement-I** : The T-wave represents the return of the ventricles from excited to normal state.

Statement-II : The end of the T-wave marks the end of systole.

- (1) Statement-I & Statement-II both are correct.
- (2) Statement-I is correct & Statement-II is incorrect.
- (3) Statement-I is incorrect & Statement-II is correct.
- (4) Both Statement-I & Statement-II is incorrect.

170. **Statement-I** : Blood is most commonly used body fluid by most of the higher organism for transport of substances.

Statement-II : Blood consist of fluid matrix (plasma) and formed elements.

- (1) Statement-I and statement-II both are true
- (2) Statement-I is true and statement-II is false
- (3) Statement-I is false and statement-II is true
- (4) Statement-I and statement-II both are false

171. **Statement-I** : In open circulatory system, blood pump by heart passes through large vessels into open spaces.

Statement-II : Open circulatory system present in Arthropoda, Mollusca & all chordates.

- (1) Statement-I and statement-II both are true
- (2) Statement-I is true and statement-II is false
- (3) Statement-I is false and statement-II is true
- (4) Statement-I and statement-II both are false

172. Which nodal musculature generate the maximum number of action potentials :

- (1) SA node
- (2) AV node
- (3) Bundle of His
- (4) Purkinje fibres

173. If the total period of one cardiac cycle is 0.8 second then, the complete cardiac diastole is _____, atrial systole is _____ and ventricular systole is :

- (1) 0.1 second; 0.3 second; 0.4 second
- (2) 0.1 second; 0.4 second; 0.3 second
- (3) 0.4 second; 0.1 second; 0.3 second
- (4) 0.3 second; 0.1 second; 0.4 second

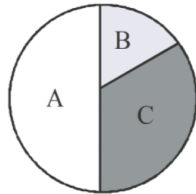
174. Which the following options **correctly** represent the blood flow pathway in pulmonary circulation?

- (1) Right ventricle $\xrightarrow[\text{Pulmonary Vein}]{\text{Pulmonary Artery}}$ Lungs
- (2) Left ventricle $\xrightarrow{\text{Aorta}}$ Body $\xrightarrow{\text{Vena Cava}}$ Right Atrium
- (3) Right ventricle $\xrightarrow[\text{Pulmonary artery}]{\text{Pulmonary Vein}}$ Lungs
- (4) Right ventricle $\xrightarrow[\text{Pulmonary Vein}]{\text{Pulmonary Artery}}$ Lungs

175. Which blood group have maximum antigen ?

- (1) AB^- (2) O^-
(3) AB^+ (4) All of these

176. In the given figure the durations of the events of the cardiac cycle are given. Identify these events and select the correct option :



	A	B	C
(1)	Auricular Systole	Joint diastole	Ventricular systole
(2)	Ventricular systole	Joint diastole	Auricular Systole
(3)	Ventricular systole	Auricular Systole	Joint diastole
(4)	Joint diastole	Auricular Systole	Ventricular systole

177. Which of the following is the correct match of supply of blood of organ and its blood ?

	Blood	Organ	Blood Vessel
(1)	Deoxygenated	Heart Muscles	Coronary artery
(2)	Oxygenated	Lungs	Pulmonary artery
(3)	Deoxygenated	Small intestine	Hepatic vein
(4)	Oxygenated	Liver	Hepatic artery

178. Clot is made up of

- (1) Fibrin network only
(2) Damaged formed elements only
(3) Damaged formed elements trapped in fibrin
(4) Damaged platelets only

179. Milk protein is acted upon by a gastric enzyme in infant mammals named :-

- (1) Rennin
(2) Caesinogen
(3) Pepsinogen
(4) Pepsin

180. Longitudinal muscular folds of inner wall of stomach are :-

- (1) Ampullae of Vater
(2) Rugae
(3) Fissures
(4) Villi/Kerckring's folds

181. Pro-carboxypeptidase, Chymotrypsinogen, Ptyalin, Amylopsin, Enterokinase, Nuclease, Amino peptidase
How many of the following enzymes are part of the pancreatic juice?

- (1) Four
(2) Five
(3) Three
(4) Six

182. How many double circulations are normally completed by the human heart in one minute :

- (1) 8
(2) 10
(3) 36
(4) 72

183. Which among the following is correct during each cardiac cycle?

- (1) The volume of blood pumped out by the right and left ventricles is same
- (2) The volume of blood pumped out by the right and left ventricles is different
- (3) The volume of blood received by each atrium is different
- (4) The volume of blood received by the aorta and pulmonary artery is different

184. Select the incorrect match pair

- (1) Hepatic portal vein : Transport blood from intestine to liver
- (2) Dorsal aorta : Transport O_2 and nutrients towards systemic organs
- (3) Hypophyseal portal vein : Transport blood from hypothalamus to posterior lobe of pituitary
- (4) Carotid artery : supply oxygenated blood to brain

185. Find the correct descending order to percentage proportion of leucocytes in human blood.

- (1) Neutrophils $\rightarrow$ Basophils $\rightarrow$ Lymphocytes $\rightarrow$ Acidophils (Eosinophils) $\rightarrow$ Monocytes
- (2) Neutrophils $\rightarrow$ Monocytes $\rightarrow$ Lymphocytes $\rightarrow$ Acidophils $\rightarrow$ Basophils
- (3) Neutrophils $\rightarrow$ Lymphocytes $\rightarrow$ Monocytes $\rightarrow$ Acidophils $\rightarrow$ Basophils
- (4) Neutrophils $\rightarrow$ Acidophils $\rightarrow$ Basophils $\rightarrow$ Lymphocytes $\rightarrow$ Monocytes

TG: @Chalnaayaaar
SECTION-B (BIOLOGY-II)

186. Which of the following statement is related to inheritance of flower colour in dog flower?

- (1) F_1 had phenotype that resembled either of the parent.
- (2) F_1 had phenotype that did not resemble either of two parents and was in between the two.
- (3) F_1 had phenotype that resembled both the parents.
- (4) F_1 had phenotype of combination of two parents

187. What is not true about *Drosophila melanogaster*?

- (1) Can be grown on simple synthetic medium in laboratory.
- (2) Complete lifecycle within two weeks
- (3) Male and female flies are easily distinguishable
- (4) Many hereditary variations can be studied by high power microscope only

188. A dioecious plant has same sex-determination mechanism as seen in many birds. In a cross between male plant and female plant, a seed is obtained having embryo of female sex. How would you describe the embryo, endosperm and seed coat of the seed obtained?

- (1) ZZ, ZZW, ZW
- (2) ZW, ZWW, ZW
- (3) ZZ, ZWW, ZW
- (4) ZZ, ZWW, ZZ

189. Which of the following is not an example of polygenic quantitative character?
- (1) Human height
 - (2) Human skin colour
 - (3) Flower colour in Garden Pea
 - (4) Kernal colour in wheat
190. The back flow of faecal matter from the large intestine into the small intestine is prevented by the presence of :-
- (1) epiglottis
 - (2) sphincter of oddi
 - (3) ileo-caecal valve
 - (4) gastro-oesophageal sphincter.
191. Read the following statements carefully :-
- (i) Food is one of the basic requirements of all living organisms.
 - (ii) Major component of our food are carbohydrates proteins and fats.
 - (iii) Vitamins and minerals are required in large amount.
 - (iv) Water plays an important role in metabolic processes and also prevents dehydration of the body.
- How many of the above statements are incorrect?
- (1) 1
 - (2) 2
 - (3) 3
 - (4) 4
192. In cardiac cycle, "Dub" sound is heard when
- (1) Mitral valves close
 - (2) Mitral valves open
 - (3) Semilunar valves close
 - (4) Bicuspid valve close
193. The walls of capillaries are composed of endothelium which is :-
- (1) Cuboidal epithelium
 - (2) Squamous epithelium
 - (3) Columnar epithelium
 - (4) Stratified epithelium
194. Osmotic pressure is maintained by blood colloid
- (1) Albumin
 - (2) Thrombin
 - (3) Fibrinogen
 - (4) Globulin
195. Which one of the following organ can be called as "blood bank" ?
- (1) Heart
 - (2) Liver
 - (3) Spleen
 - (4) Lungs
196. Kidney shaped (or bean shaped) nucleus is a feature of :-
- (1) Neutrophils
 - (2) Basophils
 - (3) Lymphocytes
 - (4) Monocytes
197. Persons with 'AB' blood group are called as "Universal recipients". This is due to :
- (1) Absence of antigens A and B on the surface of RBCs
 - (2) Absence of antigens A and B in plasma
 - (3) Presence of antibodies, anti-A and anti-B, on RBCs
 - (4) Absence of antibodies, anti-A and anti-B, in plasma

198. Hormone cause contraction of gall bladder :-

- (1) Secretin
- (2) Gastrin
- (3) G.I.P.
- (4) Cholecystokinin

199. Which digestive substance not present in hepatic portal vein ?

- (1) Fatty acid
- (2) Glucose
- (3) Amino acid
- (4) Fructose

200. Heart failure:-

- (a) Same as heart attack
- (b) Heart is not pumping blood effectively enough to meet the needs of the body.
- (c) It is often referred to as atherosclerosis
- (d) It is sometimes called congestive heart failure.
- (e) It occurs due to condition that affect the blood flow.
- (f) In this heart stops beating.

Incorrect statement are :-

- (1) a,b,c,f
- (2) c,d,e,f
- (3) b,c,d
- (4) a,c,e,f

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		C	NOMENCLATURE, ISOMERISM
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